Final Project EDA 4.17

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Set Up

```
# R packages
library(ipumsr)
library(tidyverse)
library(dplyr)
library(ggplot2)
```

```
ddi <- read_ipums_ddi("data/ahtus_00001.xml")
data <- read_ipums_micro(ddi)
```

Questions: Exploring the Data

What is your outcome variable(s)? How well does it measure the outcome you are interested in? How does it relate to your expectations?

Our primary outcome variable is time spent on paid work. However, we initially ran regressions with wage as the outcome variable and found statistically insignificant results.

We did several multiple linear regressions with wage as the outcome variable. We expected wages to have significant results, when explanatory variables such as gender, childcare, and an interaction term of $(gender \times childcare)$ were included. We expected there to be a larger wage difference from the $(gender \times childcare)$ interaction term. However, we noticed small and insignificant coefficients for this regression. We'd expect time spent on childcare to be greater for women than men.

Further, there is a trade-off between time spent at paid work versus time spent on childcare. So, to interrogate this idea of the time trade-off, we ran a subsequent regression with time spent on paid work as the outcome variable. This regression includes gender and childcare as explanatory variables, with the $(gender \times childcare)$ interaction term. Here, we found significant results and a clear visual relationship of more time spent on childcare meant less time spent on paid work. This upheld our expectations, as both women and men exhibited this relationship.

What are your key explanatory variables?

- Gender

- Time spent on childcare
- Time spent on unpaid domestic work

Research Questions

Is there a gender difference in how individuals allocate their time? How does the trade-off between time spent at paid work versus time spent on childcare differ by gender? Ultimately, the question of interest is how do demographics and expected responsibilities affect time use.

Questions: Data Wrangling and Transformation

What data cleaning did you do?

- Converted the time variables from minutes to hours for easy interpretation
- Removed rows with NAs
- Other than that, the diary entry data was well cleaned and coded by IPUMs! 🙂
- We focus on 2010-2018 samples, since historical data prior to then is sporadic and requires time series methods.

How did you wrangle the data?

We applied the following filters for the linear regression:

- Married with kids: HHTYPE == 1
- Child under 18: UNDER18 1 through 10
- Post-college education: EDUC == 5 or 6
- Full-time workers: FULLTIME == 1 (Only comparing men and women who work full-time)

Filtering was slightly less strict for Figures 1 through 4 (exploratory visualizations).

Are you deciding to exclude any observations? If so, why?

Yes, but this was mostly the data wrangling step. We removed any missing values or NAs. For some visualizations (Figure 2) a few of the time use variables were omitted to better depict explanatory variables, rather than activities we're less interested in: such as travel, media consumption, physical activity, etc.

Did you have to create any new variables from existing variables? If so, how and why?

Yes, when performing PCA on the time variables, we decided to combine a few variables to increase interpretability. ACT_INHOME and ACT_OUTHOME represented two types of leisure, so were combined into ACT_LEISURE. ACT_PCARE and ACT_PHYSICAL, which represented personal care and physical activity, were combined into ACT_SELFCARE since physical activity (exercise) is also a form of personal/self care.

We also converted several variables into factors, to be used as categorical variables in our visualizations and preliminary analysis. For instance, SEX (formerly coded as 1 and 2) was re-coded into a binary FEMALE categorical variable, 0 for men and 1 for female.

Data Cleaning

This is the primary data cleaning and transforming done.

Filtering for Visualizations 1-4

```
filtered_data <- data %>%
  filter(
    HHTYPE == 1, # filter to married with kids
                 # isolating a group where childcare is relevant
                 # avoid comparing people who don't face childcare constraints
    UNDER18 > 0, # filter to individuals with kids under age of 18
    AGE >= 22 & AGE <= 40, # between the ages of 22 and 40
    EDUC %in% c(5, 6), # college educated
    AGEYNGST >= 0 & AGEYNGST <= 18, # youngest child is between age 0 and 18
    SEX %in% c(1, 2), # male or female
    !is.na(AGE),
    !is.na(EDUC)
  ) %>%
  mutate(
    sex_label = ifelse(SEX == 1, "Male", "Female")
  )
```

Filtering for Visualization 5 (Regression Model)

```
# filtering and turning sex and full-time into binary variables
datareg <- filtered_data %>%

  filter(
    (EDUC == 5 | EDUC == 6), # filter to just college educated individuals
    FULLTIME == 1, # filter to just full-time workers
    WAGELM > 0, # remove non-earners
    ACT_CHCARE >= 0, # filter out missing or unclean data
    ACT_PCARE >= 0 # filter out missing or unclean data
  ) %>%
```

```
  mutate(
    across(starts_with("ACT_"), ~ .x / 60), # convert all time use vars to hours
    female = ifelse(SEX == 2, 1, 0),
    female = factor(female),
    FULLTIME = factor(FULLTIME),
    log_wage = log(WAGELM + 1) # add column for log wage in case it becomes interesting
  )


# filter to just 2010s data
data2010s <- datareg %>%
  filter(SAMPLE > 2009)
```

Codebook

https://github.com/ccaitlingo/westcoasters-finalproject/blob/main/README.md

Exploratory Data Analysis

Motivation

Figure 1: Unpaid Domestic Work by Sex

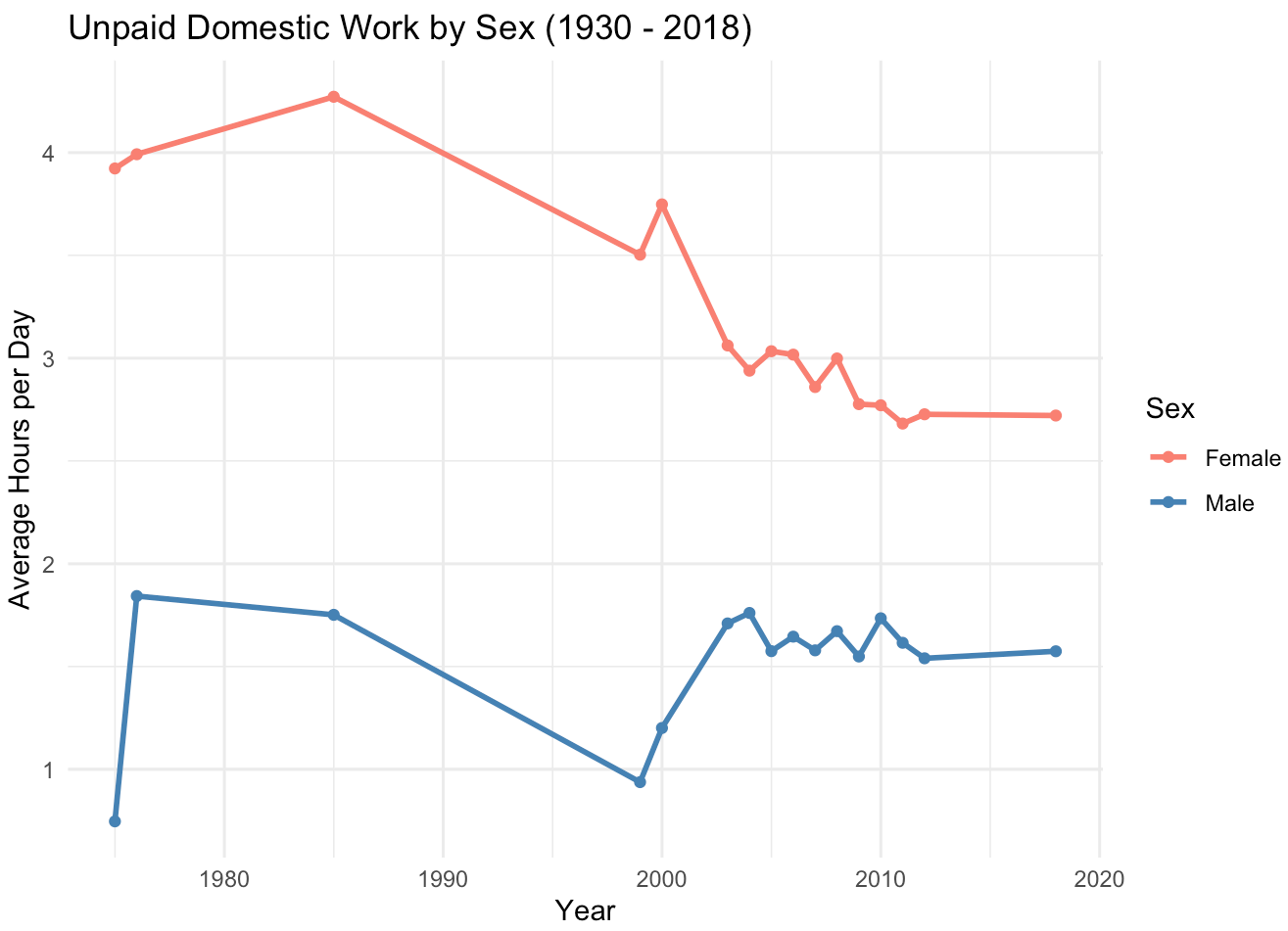
This visualization depicts the average hours per day that each sex (male or female) spends performing unpaid domestic work, from 1930 to 2018. Since our main outcome variable is time spent on **paid work**, it is also important to complement that variable by considering time spent on **unpaid work**, to get the full picture.

We can see that over time, the gap between male and female has shrunk. However, a disparity still remains: even our most current data from 2018 shows that males on average spend 1.57 hours on unpaid domestic work, while females spend 2.72 hours.

```
# Unpaid Domestic Work by Sex (1930-2018)

# Average hours by year and sex
summary_data <- filtered_data %>%
  group_by(YEAR, sex_label) %>%
  mutate(ACT_UNDOM_HRS = ACT_UNDOM / 60) %>%
  summarise(avg_domestic_hours = mean(ACT_UNDOM_HRS, na.rm = TRUE), .groups = "drop")

# Line plot
ggplot(summary_data, aes(x = YEAR, y = avg_domestic_hours, color = sex_label)) +
  geom_line(size = 1) +
  geom_point() +
  labs(
```

```
    title = "Unpaid Domestic Work by Sex (1930 – 2018)",
    x = "Year",
    y = "Average Hours per Day",
    color = "Sex"
  ) +
  theme_minimal() +
  scale_color_manual(values = c("Male" = "steelblue", "Female" = "salmon"))
```

```
Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
ℹ Please use `linewidth` instead.
```




```
# Note for 2018
# 2.72 hrs for female
# 1.57 hrs for male
```

Figure 2: Average Daily Time Use by Sex

This visualization breaks down the 24 hours in a day into the proportion of time spent on various activities, averaged across the years 2010-2018. One of the time activities is our outcome variable, time spent on **paid work**, while others include time spent performing childcare, personal care, unpaid domestic work, and leisure.

```
# How time is split by gender on an average day!

# data wrangling
df <- filtered_data %>%
  filter(SAMPLE >= 2010 ) %>%
  mutate(HHTYPE = as.numeric(as.factor(HHTYPE))) %>%
  mutate(
    Leisure = ACT_INHOME + ACT_OUTHOME
  ) %>%
  select(-"ACT_CIVIC", -"ACT_EDUCA", -"ACT_MEDIA", -"ACT_MISSING",-"ACT_INHOME", -"ACT_OU
# pivot! to have all activities listed under activity_category
df_long <- df %>%
  pivot_longer(
    cols = starts_with("ACT_") | starts_with("Leisure"),
    names_to = "activity_category",
    values_to = "minutes"
  )  %>%
  mutate(
    # Clean up the activity names (optional)
    activity_category = gsub("ACT_", "", activity_category),
    activity_category = case_when(
      activity_category == "CHCARE" ~ "Childcare",
      activity_category == "PCARE"  ~ "Personal Care",
      activity_category == "UNDOM"  ~ "Unpaid Domestic Work",
      activity_category == "WORK"   ~ "Paid Work",
      TRUE ~ activity_category
    ),
    # Convert SEX codes to labels
    SEX = as.numeric(as.factor(SEX)),
    SEX = case_when(
      SEX == 1 ~ "Male",
      SEX == 2 ~ "Female"
    )
  )

# aggregate the activities!
df_sum <- df_long %>%
  group_by(SEX, activity_category) %>%
  summarize(minutes = mean(minutes, na.rm = TRUE))


#plot
ggplot(df_sum, aes(x = SEX, y = minutes, fill = activity_category)) +
  geom_bar(stat = "identity", position = "fill") +
  labs(
    x = "Gender",
    y = "Proportion of Time Spent per Day",
    fill = "Activity",
    title = "Average Daily Time Use by Sex 2010-2018"
```

```
  ) +
  theme_minimal()
```

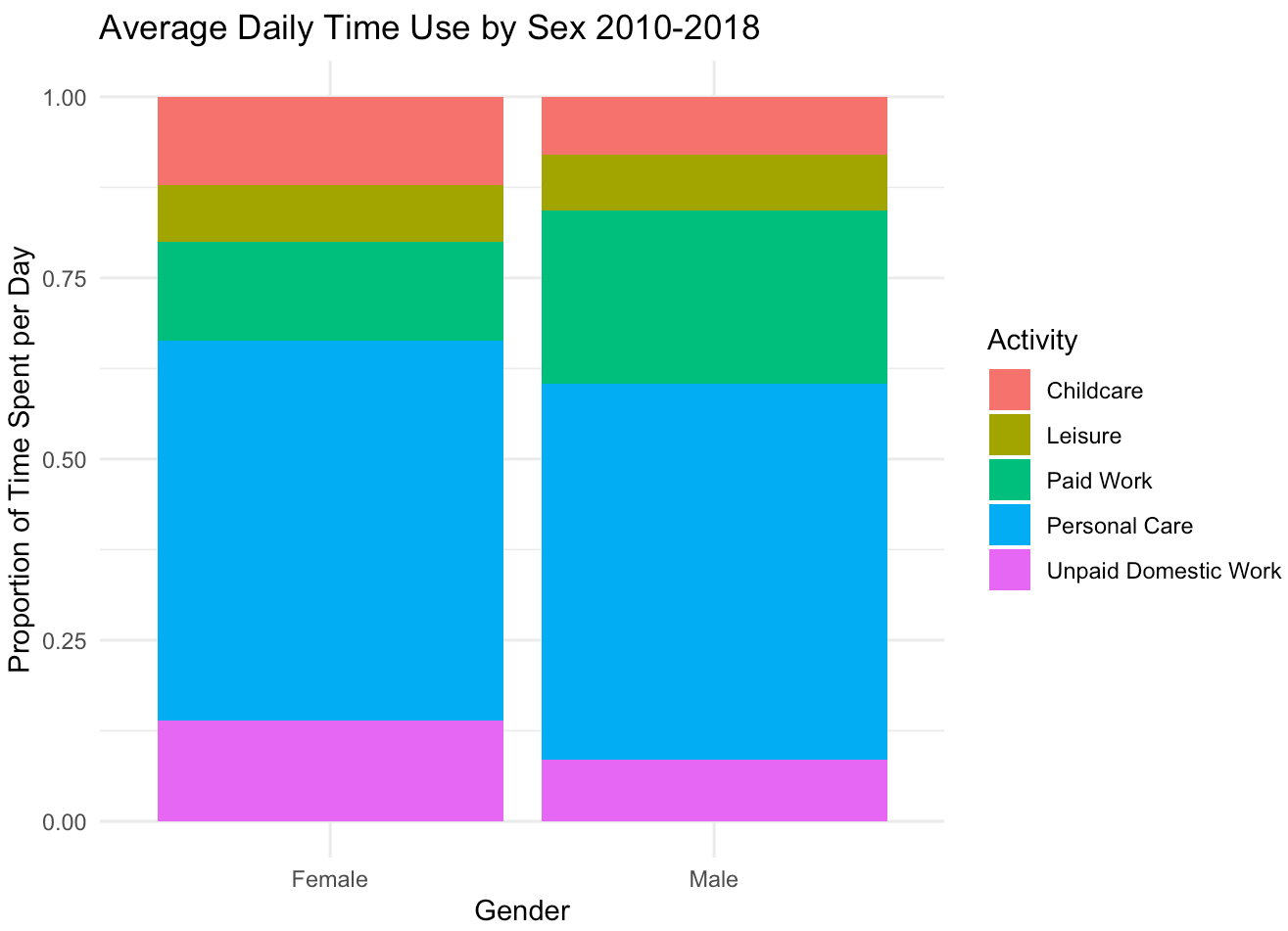


Figure 3: Clustering in 2D

This visualization uses hierarchical clustering to find groupings in **PCA space**. We discovered that data points in clusters 2 and 3, which represent the top right of the PCA space, are 69.8% female. This tells us that females are more associated with having younger children, and spending more time on unpaid domestic work and personal care!

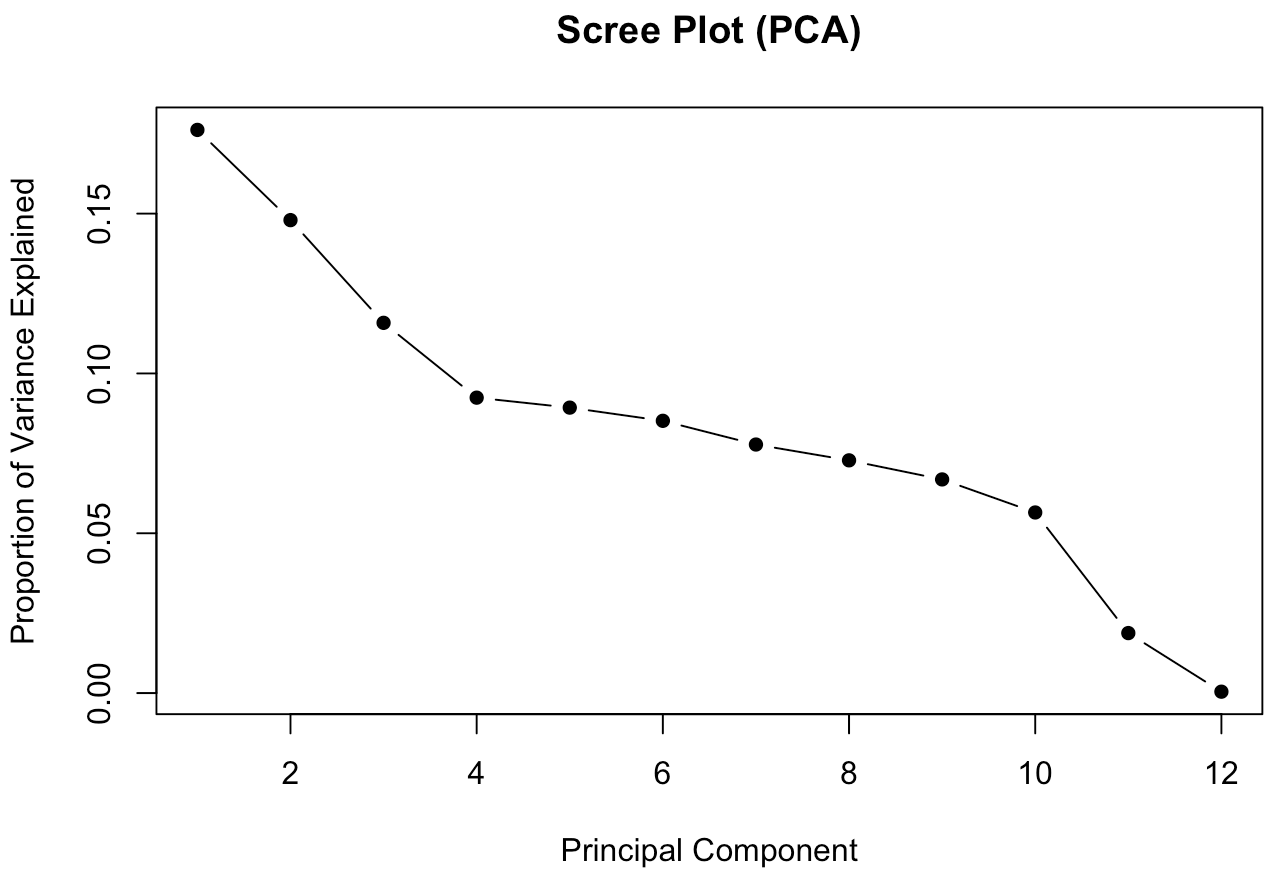
Interpreting Principal Components (PCS): See the output summary generated by the code, where the first four PCs are defined. Each PC is a vector that represents a combination of factors. The most influential factors that make up each PC are described as follows.

PCA 1: Time spent on childcare, age of the youngest child, and the amount of children under 5

PCA 2: Time spent at work, time spent on self care, and time spent doing unpaid domestic work

```
# Combine two leisure categories
filtered_data$ACT_LEISURE <- filtered_data$ACT_INHOME + filtered_data$ACT_OUTHOME
filtered_data$ACT_SELFCARE <- filtered_data$ACT_PCARE + filtered_data$ACT_PHYSICAL

# Select variables
```

```r
vars <- c(
  "ACT_WORK", "ACT_CHCARE", "ACT_EDUCA",
  "ACT_LEISURE", "ACT_MEDIA", "ACT_TRAVEL", "ACT_SELFCARE",
  "ACT_UNDOM", "ACT_CIVIC", "AGEYNGST", "UNDER5", "UNDER18"
)

# Keep only selected variables + sex
df_vars <- filtered_data[, c(vars, "SEX")]

# Drop missing values
df_vars <- na.omit(df_vars)

# Sample data (original dataset too large)
set.seed(123)
df_sample <- df_vars[sample(nrow(df_vars), 5000), ]

# Create labels
df_sample$binary_var <- ifelse(df_sample$SEX == 1, "Male", "Female")

# Standardize the variables
df_scaled <- scale(df_sample[, vars])

# Run PCA
pca_res <- prcomp(df_scaled, center = TRUE, scale. = TRUE)

# Optional scree plot (tells us to use 4 PCs for optimal analysis)
eig_vals <- pca_res$sdev^2
var_explained <- eig_vals / sum(eig_vals)

plot(var_explained, type = "b",
     xlab = "Principal Component",
     ylab = "Proportion of Variance Explained",
     main = "Scree Plot (PCA)",
     pch = 16)
```

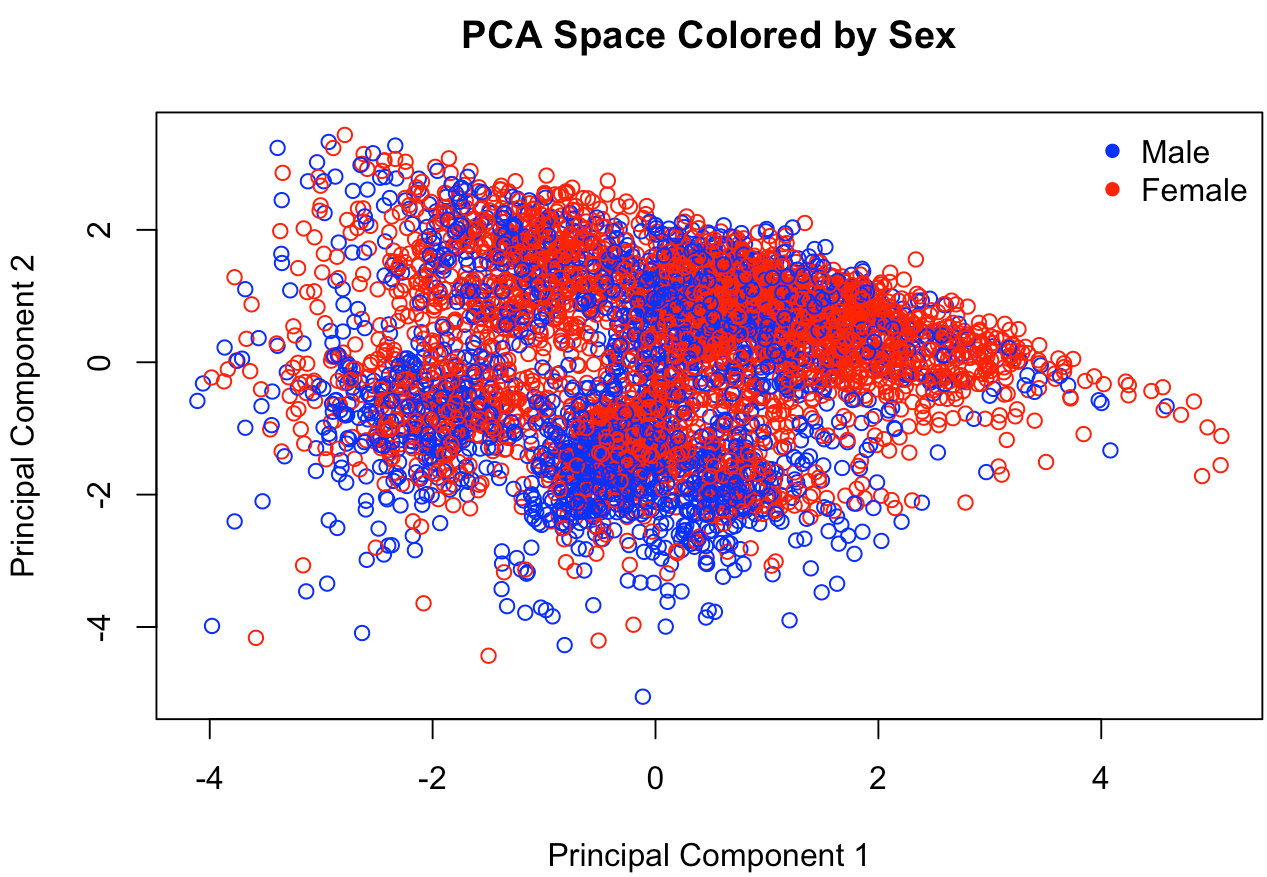



```
# View the principal components
pca_res$rotation[, 1:4]
```

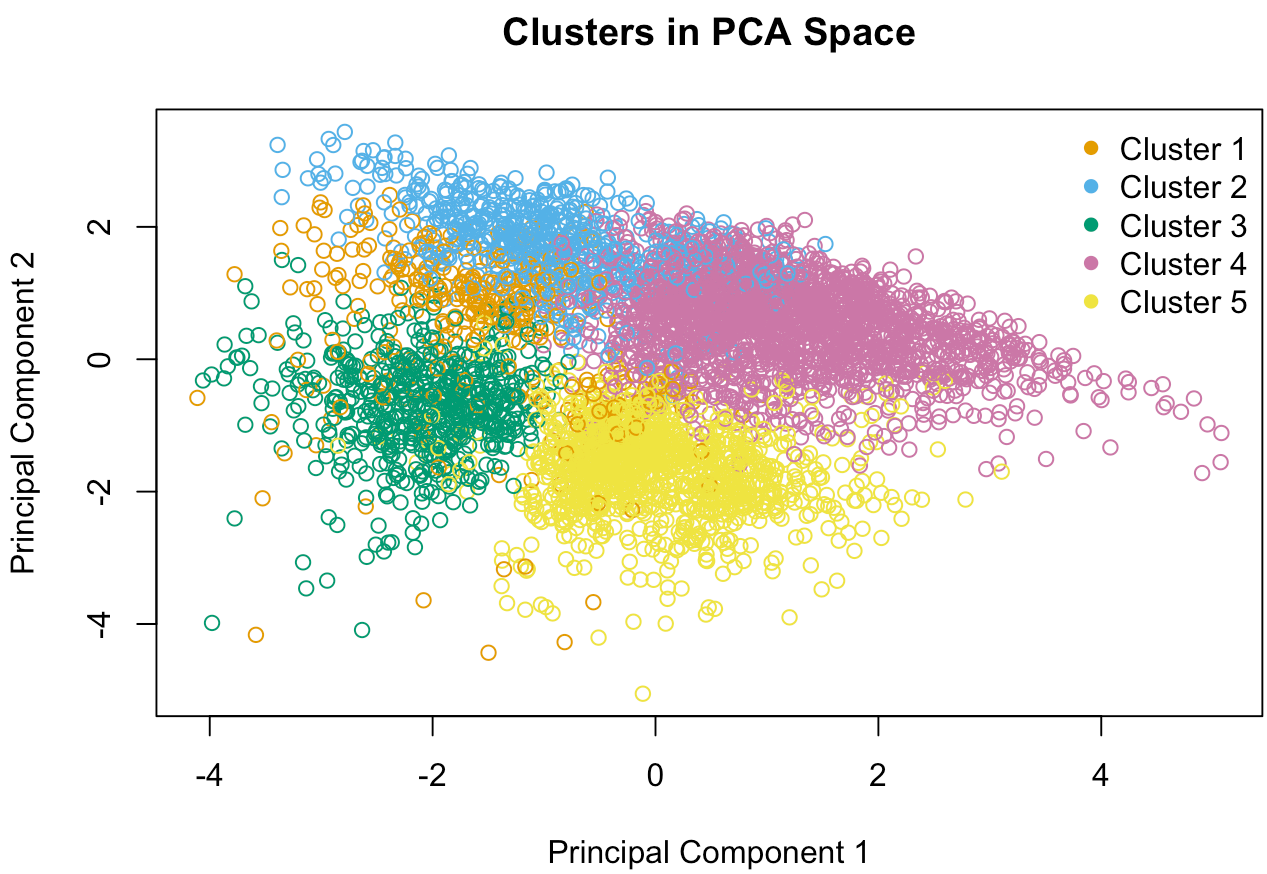
```
                     PC1         PC2         PC3         PC4
ACT_WORK     -0.25727416 -0.61638267 -0.30463198 -0.07340246
ACT_CHCARE    0.47162669  0.04833464 -0.08973658 -0.11207132
ACT_EDUCA    -0.03350391 -0.01448387 -0.01281333  0.07596707
ACT_LEISURE  -0.01925964  0.03643258  0.59254598  0.39602217
ACT_MEDIA    -0.04806055  0.33090341 -0.37894763 -0.01223683
ACT_TRAVEL   -0.14358451 -0.19605454  0.51496656  0.02069748
ACT_SELFCARE  0.02024284  0.41974261 -0.04744463  0.34480268
ACT_UNDOM     0.18011435  0.40680907  0.05396275 -0.29758501
ACT_CIVIC    -0.03531322  0.11765900  0.30204406 -0.38635598
AGEYNGST     -0.55817780  0.22476147  0.01692206 -0.22939443
UNDER5        0.56581452 -0.23833731  0.02161607  0.07749477
UNDER18       0.14543721 -0.07362807  0.20477481 -0.63523658
```

```
# Extract first 4 PCs
pca_scores <- as.data.frame(pca_res$x[, 1:4])

# Hierarchical clustering on PCA space
dist_mat <- dist(pca_scores)
hc <- hclust(dist_mat, method = "ward.D2")
```

```r
# Cut into clusters
k <- 5
clusters <- cutree(hc, k = k)

# Attach cluster + sex
pca_scores$cluster <- factor(clusters)
pca_scores$binary_var <- df_sample$binary_var

# -------------------------
# Plot 1: PCA Space Colored by Sex
# -------------------------
plot(pca_scores$PC1, pca_scores$PC2,
     col = ifelse(pca_scores$binary_var == "Male", "blue", "red"),
     pch = 1,
     xlab = "Principal Component 1",
     ylab = "Principal Component 2",
     main = "PCA Space Colored by Sex")

legend("topright",
       legend = c("Male", "Female"),
       col = c("blue", "red"),
       pch = 16,
       bty = "n")
```



```
# -------------------------
# Plot 2: Clusters in PCA Space
# -------------------------
cluster_colors <- c("#E69F00", "#56B4E9", "#009E73", "#CC79A7", "#F0E442")

plot(pca_scores$PC1, pca_scores$PC2,
     col = cluster_colors[pca_scores$cluster],
     pch = ifelse(pca_scores$binary_var == "Male", 1, 1),
     xlab = "Principal Component 1",
     ylab = "Principal Component 2",
     main = "Clusters in PCA Space")

legend("topright",
       legend = c("Cluster 1","Cluster 2","Cluster 3","Cluster 4", "Cluster 5"),
       col = cluster_colors,
       pch = 16,
       bty = "n")
```




```
# -------------------------
# Table
# -------------------------
# Counts
tab <- table(pca_scores$cluster, pca_scores$binary_var)
```

```
# Row-wise percentages (within cluster)
pct <- round(prop.table(tab, margin = 1) * 100, 1)
pct
```

```
   Female Male
 1   65.1 34.9
 2   67.3 32.7
 3   53.5 46.5
 4   68.0 32.0
 5   42.6 57.4
```

```
# Group clusters 2+3 (representing the top right of graph) and 1+4+5
grp <- ifelse(pca_scores$cluster %in% c(2,3), "C2+3", "C1+4+5")
pct_grp <- round(prop.table(table(grp, pca_scores$binary_var), 1) * 100, 1)
pct_grp
```

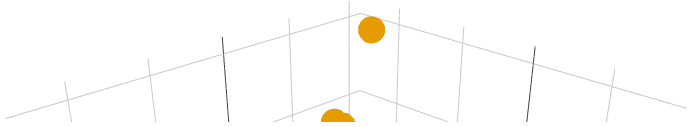
```
grp      Female Male
  C1+4+5   59.1 40.9
  C2+3     60.8 39.2
```

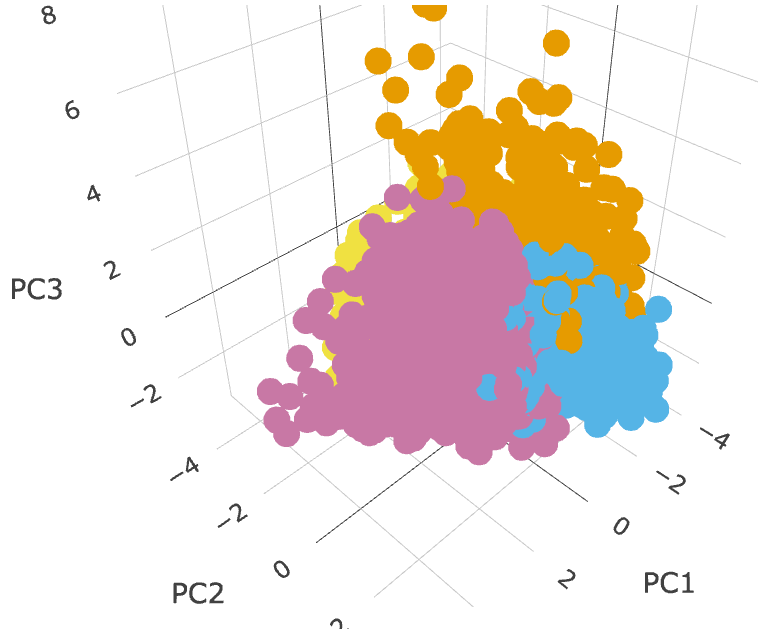
Figure 4: Clustering in 3D

This visualization shows the same clustering in PCA space, but in three dimensions instead of two. This allows us to visualize the third PC.

PC3: Time spent on leisure, time spent on travel

```
library(plotly)

# 3D PCA plot
plot_ly(
  pca_scores,
  x = ~PC1,
  y = ~PC2,
  z = ~PC3,
  color = ~as.factor(cluster),
  colors = cluster_colors,
  type = "scatter3d",
  mode = "markers"
)
```





Preliminary Analysis

Figure 5: Exploring the Research Question

In this visualization, we begin to observe the time trade-off outlined in our research question and hypothesis - there appears to be an opportunity cost associated with time spent at paid work versus time spent on childcare. There's not an apparent difference between gender, however there's a substantial split when time spent on childcare is less than 1 hour a day. While this could be noise, maybe we could interrogate this further in a regression. Otherwise, the gender split is minimal! Both exhibit a clear inverse relationship between time spent on paid work relative to time spent on childcare.

```
data2010s %>%
  mutate(chcare_bin = ntile(ACT_CHCARE, 5)) %>%
  group_by(chcare_bin, female) %>%
  summarise(
    avg_wage = mean(ACT_WORK, na.rm = TRUE),
    avg_chcare = mean(ACT_CHCARE)
  ) %>%
  ggplot(aes(x = avg_chcare, y = avg_wage, color = female)) +
  geom_line() +
  geom_point() +
    labs(
      title = "Time Spent on Paid Work vs. Time Spent on Childcare by Gender",
      subtitle = "In the 2010s for Individuals who are College-Educated and Married with
      x = "Average Hours of Childcare per Day",
      y = "Average Hours of Paid Work per Day"
    )
```

Time Spent on Paid Work vs. Time Spent on Childcare by Gender
In the 2010s for Individuals who are College-Educated and Married with Kids Under 18
Average Hours of Paid Work per Day
5
4
3
2
0
1
2
3
4
5
Average Hours of Childcare per Day
female
0
1